# Signs and Checking

SAT Math · Foundations to Mastery · Week of 27 September

Name: _____ Due: _____

**How to use this sheet**

This is four short sittings, not one long one. Four half-hours across the week will do more for you than one long Saturday, and you are far more likely to finish it.

- **Part A is no calculator.** None. It is 25 minutes and it is the most important part.
- **Write every substitution in brackets.** Every one. This is the habit that carries the most marks.
- When you get one wrong, write it in your error log: what you did, and what you should have done.

| Sitting | Part | What it drills | Minutes |
|---|---|---|---|
| Sunday | A | Signed arithmetic, no calculator | 25 |
| Tuesday | B, C | Substitution, and the four traps | 30 |
| Thursday | D | "Could that be true?" | 20 |
| Saturday | E | Six mixed questions, timed | 25 |

## Part A — Signed arithmetic

*No calculator. 25 minutes.*

SUNDAY

Work down the list. If you are unsure of a sign, draw a number line in the margin.

1. $7-(-1)=$ _____
2. $-5-3=$ _____
3. $-4-(-9)=$ _____
4. $6-(-6)=$ _____
5. $(-3)(-7)=$ _____
6. $(-2)^4=$ _____
7. $(-2)^3=$ _____
8. $-2^4=$ _____
9. $-(-11)=$ _____
10. $\dfrac{-20}{-4}=$ _____
11. $12-(-8)+(-5)=$ _____
12. $-(3-8)=$ _____
13. $(-1)(-1)(-1)=$ _____
14. $5-2(-4)=$ _____
15. $(-6)^2-6^2=$ _____

Expand and simplify.

16. $-3(x-4)$ _____
17. $-(2x-7)$ _____
18. $(x-5)(x+2)$ _____
19. $(x+4)(x-6)$ _____
20. $-2(3-x)+4$ _____

## Part B — Substitution, in brackets

*Calculator allowed. 15 minutes.*

TUESDAY

Write the substitution in brackets *before* you work anything out. Like this:

$$f(x) = 3x^2 - x, \qquad f(-2) = 3(-2)^2 - (-2) = 3(4) + 2 = 14$$

You lost marks on the diagnostic by skipping that middle step. Do not skip it here.

1. $f(x) = 3x^2 - x$. Find $f(-1)$.
2. $g(x) = x^2 - 5x$. Find $g(-3)$.
3. $h(x) = -x^2 + 4$. Find $h(-5)$.
4. $p(x) = 2x^2 + 3x - 1$. Find $p(-4)$.
5. $q(x) = 5 - x^2$. Find $q(-6)$.
6. $r(x) = x^3 + 2x$. Find $r(-2)$.
7. $s(x) = -2x - x^2$. Find $s(-3)$.
8. $f(x) = 3x^2 - x$. Find $f(-2)$.

## Part C — The four traps

*Calculator allowed. 15 minutes.*

TUESDAY

These are the four places a sign went wrong on your diagnostic, with new numbers.

1. A line passes through $(3,-2)$ and $(7,6)$. What is its slope?
   A) 1 B) 2 C) 4 D) 8

2. A line passes through $(-1,5)$ and $(3,-3)$. What is its slope?
   A) $-4$ B) $-2$ C) 2 D) 4

3. Which expression is equivalent to $(x+6)(x-4)$?
   A) $x^2 - 24$ B) $x^2 - 2x - 24$ C) $x^2 + 2x - 24$ D) $x^2 + 10x - 24$

4. Which expression is equivalent to $(x-7)(x+3)$?
   A) $x^2 - 4x - 21$ B) $x^2 + 4x - 21$ C) $x^2 - 21$ D) $x^2 - 10x - 21$

5. Which inequality gives all values of $x$ for which $7 - 3x \geq 19$?
   A) $x \geq -4$ B) $x \leq -4$ C) $x \geq 4$ D) $x \leq 4$

6. Which inequality gives all values of $x$ for which $4 - 5x < 24$?
   A) $x < -4$ B) $x > -4$ C) $x < 4$ D) $x > 4$

7. The graph of $y = (x-2)^2 - 9$ crosses the $x$-axis at two points. What are their $x$-coordinates?
   A) $-1$ and 5 B) 1 and 5 C) $-5$ and 1 D) 2 and 9

8. A line is given by $5x - 2y = 14$. What is the $y$-coordinate of its $y$-intercept?
   Answer: _____ *(write your answer)*

## Part D — Could that be true?

*No calculator. 20 minutes.*

THURSDAY

For each one, a student has given an answer. **Do not work the question out.** Say whether the answer could possibly be right, and in one sentence say how you know.

This is the most valuable part of the sheet. Five of the marks you lost on the diagnostic would have survived thirty seconds of this.

1. A right triangle has legs of length 9 and 12. A student says the hypotenuse is 0.013.

2. Of 45 students who walk to school, 26 are in Grade 11. One walker is chosen at random. A student says the probability they are in Grade 11 is 1.

3. Solve $5 - 2x \geq 11$. A student says $x \geq 3$. Test $x = 4$.

4. A jacket costs \$260 and is reduced by 15%. A student says the sale price is \$299.

5. A town of 250 people grows by 8% a year. A student says that after 3 years there are 472.

6. A cylinder has radius 3 and height 10. A student says the volume is $30\pi$.

## Part E — Mixed, timed

*Calculator allowed. 12 minutes, timed.*

SATURDAY

Set a timer for 12 minutes. Answer every one — there is no penalty for a wrong answer, so a blank is a mark thrown away. Then check each answer by substituting it back.

1. What value of $x$ satisfies $3(x-4) = x+2$?
   A) 5 B) 7 C) $-5$ D) $-7$

2. A line passes through $(-2, 3)$ and $(2, -5)$. What is its slope?
   A) $-4$ B) $-2$ C) 2 D) 4

3. The function $f$ is defined by $f(x) = x^2 - 4x$. What is the value of $f(-3)$?
   Answer: _____ *(write your answer)*

4. Which inequality gives all values of $x$ for which $8 - 4x > 20$?
   A) $x > -3$ B) $x < -3$ C) $x > 3$ D) $x < 3$

5. Which expression is equivalent to $(x-5)(x+2)$?
   A) $x^2-10$ B) $x^2+3x-10$ C) $x^2-3x-10$ D) $x^2-7x-10$

6. The graph of $y=(x+1)^2-16$ crosses the $x$-axis at two points. What is the greater of the two $x$-coordinates?
   Answer: _____ *(write your answer)*

## Error log

*Fill this in as you go.*

| Date | Q | What I did | What I should have done |
|---|---|---|---|
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*Bring this sheet and your error log to the next session.*